

ADA, OK, USA

WMO: 722044

Lat: 34.804N

Lon: 96.671W

Elev: 1016

StdP: 14.16

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53930

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 17.8	(c) 20.9	(d) 4.6	(e) 7.2	(f) 20.0	(g) 9.7	(h) 9.4	(i) 30.9	(j) 23.9	(k) 46.1	(l) 20.9	(m) 47.2	(n) 6.2	(o) 340	(p) 0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.9	(c) 99.7	(d) 74.2	(e) 97.0	(f) 74.3	(g) 93.3	(h) 74.0	(i) 78.5	(j) 90.4	(k) 77.6	(l) 89.7	(m) 76.7	(n) 88.5	(o) 8.6	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	137.8	83.7	73.4	129.1	81.7	73.0	127.3	81.4	42.7	91.2	41.8	89.0	40.7	89.2	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	19.0	17.4	8.0	102.7	5.1	3.8	4.4	105.4	1.4	107.7	-1.4	109.8	-5.1	112.6
			7.5	80.0	4.9	1.5	4.0	81.1	1.2	81.9	-1.6	82.8	-5.1	83.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.4	41.1	44.9	54.1	61.5	69.0	78.5	82.2	81.8	74.9	63.6	52.8	42.9
	DBStd	16.77	10.58	11.28	10.14	8.34	7.42	4.81	5.22	5.68	6.39	8.57	9.58	9.53
	HDD50	952	311	211	73	13	2	0	0	0	0	5	77	260
	HDD65	3070	742	565	356	158	46	0	0	0	7	131	378	687
	CDD50	5468	35	69	201	358	592	856	999	984	747	427	161	39
	CDD65	2111	0	3	19	53	171	406	534	519	304	89	12	1
	CDH74	22467	7	29	128	427	1398	4251	6370	6072	2914	764	104	3
	CDH80	10474	0	4	23	92	424	1913	3363	3181	1243	216	15	0
Wind	WSAvg	7.9	8.0	8.6	9.5	9.7	8.4	7.9	6.4	6.2	6.4	7.4	8.6	7.9
Precipitation	PrecAvg	40.0	1.8	2.1	3.2	3.7	5.3	4.2	2.8	3.1	4.6	4.0	2.8	2.3
	PrecMax	64.0	7.4	6.9	8.0	11.4	14.5	10.8	8.5	11.1	12.5	12.4	6.9	6.6
	PrecMin	23.5	0.0	0.0	0.3	0.4	0.2	0.8	0.0	0.0	0.5	0.0	0.3	0.2
	PrecStd	8.9	1.4	1.6	1.9	2.2	3.1	2.3	2.2	2.2	2.9	3.3	1.9	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.3	78.9	82.2	88.3	92.6	97.5	104.1	104.5	97.1	90.5	81.9	72.8
		MCWB	55.1	59.6	63.3	65.6	71.8	72.3	73.8	73.2	73.6	70.5	64.6	59.1
	2%	DB	68.0	72.2	77.3	81.7	88.2	94.7	99.7	100.1	92.9	85.8	77.1	67.8
		MCWB	54.1	57.8	62.2	65.2	71.9	73.6	74.3	74.1	73.2	70.6	61.1	56.4
	5%	DB	63.3	66.5	73.0	79.1	84.2	91.1	97.1	97.2	90.4	81.7	72.4	63.0
		MCWB	51.7	56.6	60.4	64.1	70.5	73.8	74.5	74.8	72.5	66.9	60.4	53.6
	10%	DB	58.5	63.0	70.3	75.1	81.5	90.0	93.3	93.2	88.0	78.8	68.2	58.6
		MCWB	48.3	53.8	59.1	62.3	69.3	73.8	74.3	75.3	71.8	65.4	58.5	51.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.4	64.2	67.9	71.6	76.1	79.9	80.1	80.4	77.8	75.2	68.3	65.1
		MCDB	66.2	70.5	76.7	78.8	85.7	90.7	91.2	92.6	90.7	84.2	77.3	69.3
	2%	WB	57.5	61.5	65.7	69.4	74.2	77.8	78.3	78.6	76.3	73.5	65.8	60.5
		MCDB	64.0	67.2	73.1	76.5	83.5	88.3	90.5	90.8	87.7	80.8	73.3	64.1
	5%	WB	53.0	58.7	63.7	67.8	72.8	76.7	77.3	77.6	75.0	71.4	62.8	55.4
		MCDB	60.8	65.4	69.3	75.0	81.7	87.0	89.7	90.3	85.4	78.2	68.9	59.6
	10%	WB	49.1	54.5	61.5	65.9	71.3	75.5	76.4	76.6	73.8	69.0	59.9	51.5
		MCDB	57.2	62.4	68.3	73.1	79.4	85.5	88.7	88.7	83.3	74.7	66.4	56.1
Mean Daily Temperature Range	5% DB	MDBR	20.0	19.9	20.8	20.8	18.4	18.8	19.9	19.9	20.9	21.6	21.3	18.6
		MCDBR	27.1	26.3	23.8	23.4	21.5	20.9	23.1	23.2	23.8	23.7	25.3	25.8
		MCWBR	16.4	15.7	13.2	12.5	9.8	7.4	6.5	6.5	9.3	11.5	13.8	17.1
	5% WB	MCDBR	24.7	22.0	19.0	19.5	18.3	18.2	19.9	20.0	19.8	18.8	22.2	21.1
		MCWBR	17.8	15.4	11.7	13.0	9.7	8.5	7.7	7.5	8.6	10.4	15.8	18.3
Clear-Sky Solar Irradiance	taub		0.300	0.307	0.346	0.376	0.410	0.423	0.429	0.434	0.388	0.345	0.320	0.300
	taud		2.503	2.501	2.417	2.346	2.316	2.316	2.341	2.294	2.382	2.494	2.501	2.509
	Ebn at Noon		289	299	294	288	279	273	271	268	276	281	279	282
	Edh at Noon		26	30	35	39	41	41	40	41	36	30	26	25
All-Sky Solar Radiation	RadAvg		862	1036	1368	1683	1814	2075	2121	1931	1600	1224	914	738
	RadStd		89	156	91	127	188	219	118	117	156	171	115	90

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page